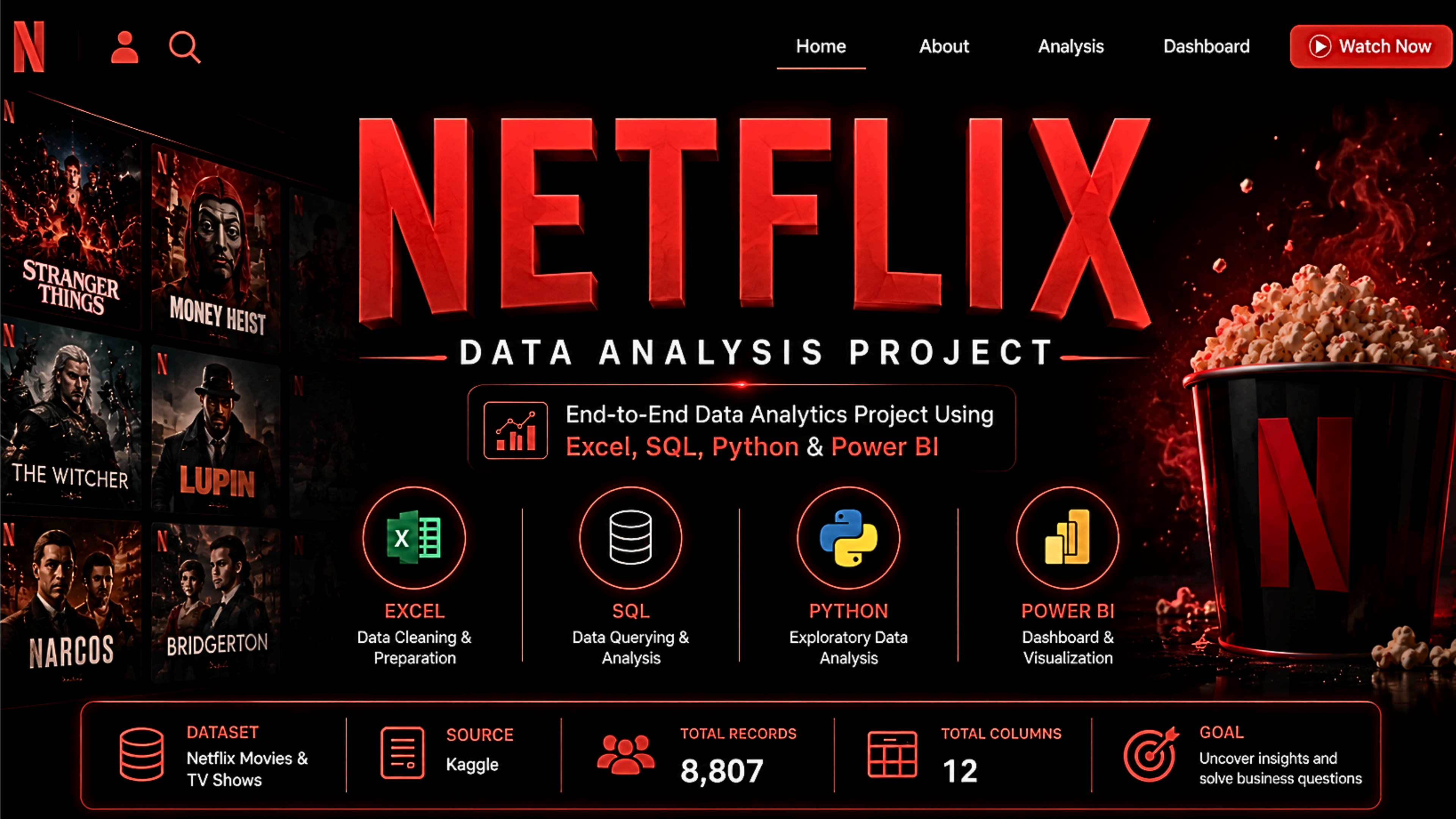




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NETFLIX

DATA ANALYSIS PROJECT



End-to-End Data Analytics Project Using
Excel, SQL, Python & Power BI



EXCEL

Data Cleaning &
Preparation



SQL

Data Querying &
Analysis



PYTHON

Exploratory Data
Analysis



POWER BI

Dashboard &
Visualization



DATASET

Netflix Movies &
TV Shows



SOURCE

Kaggle



TOTAL RECORDS

8,807



TOTAL COLUMNS

12



GOAL

Uncover insights and
solve business questions

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PROBLEM OVERVIEW

The objective of this project is to analyze Netflix's content library and identify trends in content type, ratings, countries, and release years using Excel, SQL, Python, and Power BI.

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DATASET OVERVIEW

Dataset: Netflix Movies and TV Shows

Total Records: (your dataset count)

Source: Kaggle

Tools Used:

- Excel
 - SQL
 - Python (Pandas, Matplotlib)
 - Power BI
-

EXCEL DATA CLEANING & PREPARATION

The raw Netflix dataset was cleaned and prepared in Excel to ensure data quality, consistency, and reliability for further analysis.

DATASET OVERVIEW

 Dataset	Netflix Movies & TV Shows
 Source	Kaggle
 Total Records	8,807
 Total Columns	12
 Tool Used	Microsoft Excel 

DATA CLEANING TASKS PERFORMED

- ✓ Removed duplicate records
- ✓ Identified and handled missing values
- ✓ Standardized date formats in 'Date Added' column
- ✓ Created a new calculated column: Content Age
- ✓ Converted dataset into an Excel Table
- ✓ Verified data consistency and correctness

MISSING VALUE ANALYSIS

Column Name	Missing Values
Director	2,634
Cast	825
Country	831
Date Added	10
Rating	4

CONTENT AGE CALCULATION

A new column 'Content Age' was created to find out how old each content is.

Formula Used:

`=YEAR(TODAY())-[@release_year]`

CLEANED DATASET PREVIEW

A	B	C	D	E	F	G	H
show_id	type	title	country	release_year	rating	date_added	content_age
s1	Movie	Dick Johnson Is Dead	United States	2020	PG-13	9/25/2021	5
s2	TV Show	Blood & Water	South Africa	2020	TV-MA	9/24/2021	5
s3	Movie	Ganglands	France	2021	TV-MA	9/24/2021	4
s4	TV Show	Jailbirds New Orleans	United States	2021	TV-MA	9/24/2021	4
s5	Movie	Kota Factory	India	2019	TV-PG	9/24/2021	6
...	...	...	...	...	...	...	...
s8803	Movie	Zinzana	Syria	2015	TV-14	3/11/2018	10
s8804	Movie	Zubaan	India	2015	TV-14	3/2/2019	10
s8805	Movie	Zulu Man in Japan	Japan	2019	TV-14	8/16/2019	6
s8806	TV Show	Zumbo's Just Desserts	Australia	2016	TV-PG	2/18/2019	9
s8807	Movie	Zwara	South Africa	2018	TV-MA	7/15/2021	7

* Sample of cleaned dataset in Excel



EXCEL ANALYSIS – KEY INSIGHTS

Pivot Table Analysis of Netflix Dataset



1. CONTENT RELEASE TREND OVER THE YEARS

Number of Titles Added to Netflix by Year

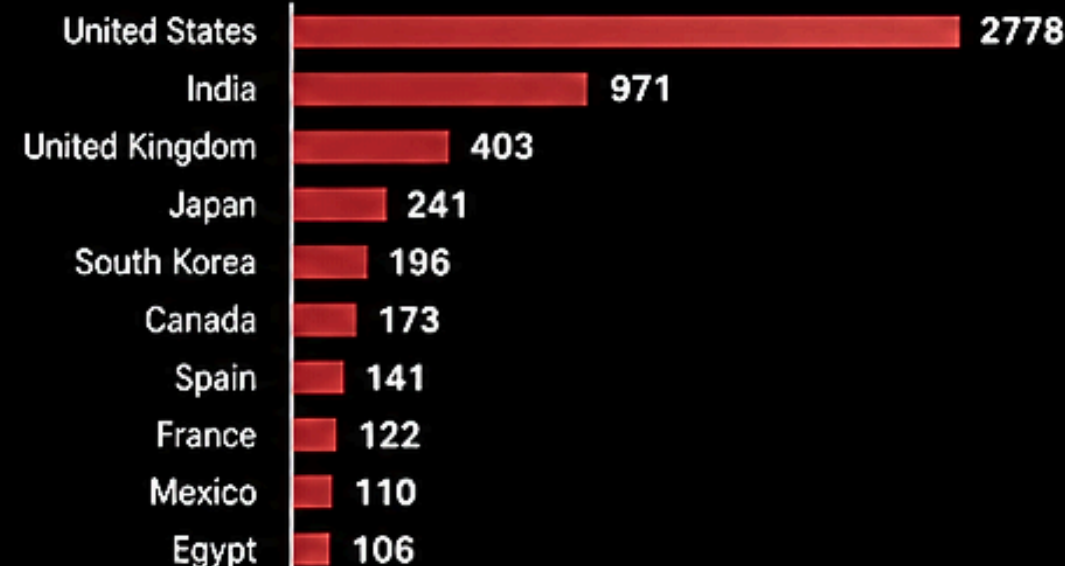


INSIGHT: Netflix content additions increased significantly after 2015, peaking around 2018 and slightly declining after.



2. TOP 10 COUNTRIES BY NUMBER OF TITLES

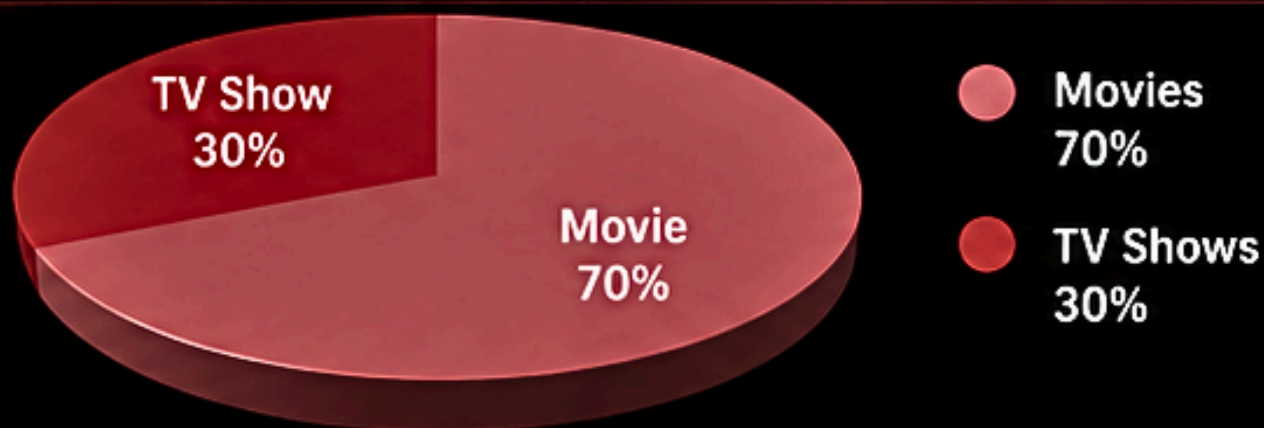
Country	Count of title
United States	2778
India	971
United Kingdom	403
Japan	241
South Korea	196
Canada	173
Spain	141
France	122
Mexico	110
Egypt	106
Grand Total	5241



INSIGHT: The United States leads with the highest number of titles, followed by India and the United Kingdom.



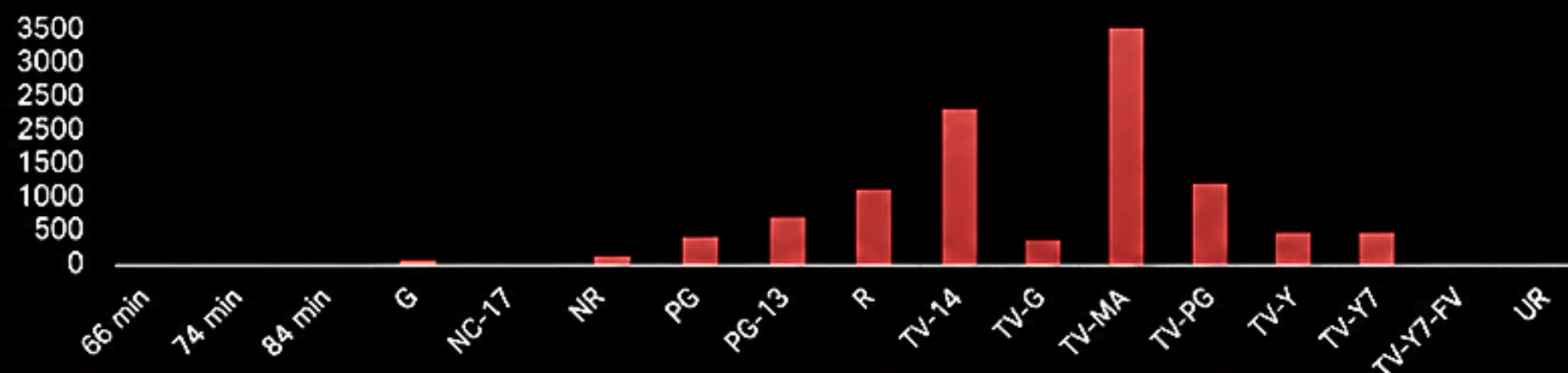
3. MOVIES VS TV SHOWS



INSIGHT: Movies make up the majority of Netflix content with 70%, while TV Shows account for 30%.



4. RATINGS DISTRIBUTION



INSIGHT: TV-MA has the highest number of titles, followed by TV-14. Most content is targeted towards mature and teen audiences.



KEY TAKEAWAYS



Movies dominate Netflix's catalog (70%).



The United States is the leading content producer.



Netflix expanded rapidly after 2015, peaking in 2018.



Content caters to a wide range of age groups and ratings.



SQL PROJECT OVERVIEW & DATA STRUCTURE



DATASET INFORMATION

Dataset Name	Netflix Dataset
Source	Kaggle
File Name	netflix_titles.csv
Records	7,864
Columns	12
Description	This dataset contains information about movies and TV shows available on Netflix, including details such as cast, director, country, rating, release year, duration and more.
Objective	To perform data analysis using SQL queries and extract meaningful insights from the Netflix dataset.



TABLE INFORMATION

Database Name	netflix_db
Table Name	netflix
Storage Engine	InnoDB
Total Rows	7,864
Created On	-

SQL SNIPPET (CREATE TABLE)

```
1 CREATE TABLE netflix (  
2   show_id      VARCHAR(20),  
3   type         VARCHAR(20),  
4   title        VARCHAR(300),  
5   director     TEXT,  
6   cast         TEXT,  
7   country      TEXT,  
8   date_added   DATE,  
9   release_year INT,  
10  rating       VARCHAR(20),  
11  duration     VARCHAR(50),  
12  listed_in    TEXT  
13 );
```



SAMPLE DATA PREVIEW (TOP 5 ROWS)

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	-	United States	2021-09-25	2020	PG-13	90 min	Documentaries
s2	TV Show	Blood & Water	-	Ama Qamata, Khosi Ngema	South Africa	2021-09-24	2021	TV-MA	1 Season	TV Dramas
s3	Movie	Jailbirds New Orleans	-	-	United States	2021-09-24	2021	TV-MA	1h 21 min	Docuseries
s4	TV Show	King's Affection	Song Hyun-wook	Park Eun-bin, Rowoon	South Korea	2021-09-24	2021	TV-14	1 Season	International TV Shows
s5	Movie	The Starling	Theodore Melfi	Melissa McCarthy, Chris O'Dowd	United States	2021-09-24	2021	PG-13	1h 44 min	Comedies, Dramas



DATA TYPES USED

show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
VARCHAR(20)	VARCHAR(20)	VARCHAR(300)	TEXT	TEXT	TEXT	DATE	INT	VARCHAR(20)	VARCHAR(50)	TEXT
Unique ID for each movie or TV show	Type of content: Movie or TV Show	Title of the movie or TV show	Director of the movie or show	Cast / actors involved	Country where the content was produced	Date when the content was added on Netflix	Year the content was released	Content rating (e.g. TV-MA, PG-13)	Duration or number of seasons	Genre(s) of the content

N SQL ANALYSIS & BUSINESS INSIGHTS

Analysis Performed Using MySQL Workbench

1. MOVIES VS TV SHOWS

```
1 SELECT type,
2 COUNT(*) AS total_titles
3 FROM netflix
4 GROUP BY type;
```

RESULT

type	total_titles
Movie	5676
TV Show	2188

 **INSIGHT:** Movies account for approximately 70% of Netflix content, while TV Shows account for 30%.

2. TOP 10 COUNTRIES BY NUMBER OF TITLES

```
1 SELECT country,
2 COUNT(*) AS total_titles
3 FROM netflix
4 GROUP BY country
5 ORDER BY total_titles DESC
6 LIMIT 10;
```

RESULT

country	total_titles
United States	2766
India	971
United Kingdom	403
Japan	241
South Korea	196
Canada	173
Spain	141
France	121
Mexico	110
Egypt	106

 **INSIGHT:** The United States leads with the highest number of titles, followed by India and the United Kingdom.

3. RATINGS DISTRIBUTION

```
1 SELECT rating,
2 COUNT(*) AS total_titles
3 FROM netflix
4 GROUP BY rating
5 ORDER BY total_titles DESC;
```

RESULT

rating	total_titles	rating	total_titles
TV-MA	2897	TV-G	183
TV-14	1902	NR	78
R	785	G	41
TV-PG	748	TV-Y7-FV	4
PG-13	481	NC-17	3
PG	280	UR	3
TV-Y7	232	74 min	1
TV-Y	221	84 min	1
		66 min	1

 **INSIGHT:** TV-MA is the most common rating, followed by TV-14. Most content is targeted towards mature and teen audiences.

4. CONTENT RELEASE TREND OVER THE YEARS

```
1 SELECT release_year,
2 COUNT(*) AS total_titles
3 FROM netflix
4 GROUP BY release_year
5 GROUP BY release_year;
```

RESULT

release_year	total_titles	release_year	total_titles
2005	77	2013	264
2006	94	2014	324
2007	87	2015	505
2008	131	2016	814
2009	141	2017	949
2010	174	2018	1029
2011	178	2019	909
2012	214	2020	851
		2021	381

 **INSIGHT:** Netflix content increased rapidly after 2015, peaking in 2018 and slightly declining after.

5. LATEST CONTENT ADDED

```
1 SELECT title,
2 release_year
3 FROM netflix
4 ORDER BY release_year DESC;
```

RESULT

title	release_year
The Netflix Afterparty	2021
Dad Stop Embarrassing Me!	2021
Blood & Water	2021
Kota Factory	2021
Secret Magic Control Agency	2021
The Great British Baking Show	2021
The Starling	2021
...	...

 **INSIGHT:** Netflix continuously expands its library with recent content releases.

KEY FINDINGS

- ✓ Movies account for 70% of Netflix content.
- ✓ The United States contributes the highest number of titles (2,766), followed by India (971) and the United Kingdom (403).
- ✓ TV-MA is the most common rating with 2,897 titles, followed by TV-14 with 1,902 titles.
- ✓ Netflix content growth accelerated significantly after 2015, peaking in 2018 with 1,029 titles.
- ✓ Netflix continues to expand its library with a strong focus on recent content additions.

NETFLIX
CONTENT ANALYTICS

Filters

country

All

type

All

rating

All

date_added

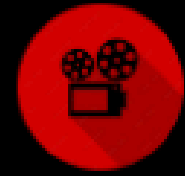
01-01-2008

25-09-2021

release_year

1942

2021



Total Movies

6K



Total Countries

743



Total Content

8K



Total TV Shows

2K

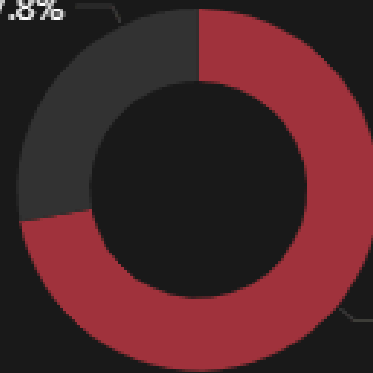


Latest Year

2021

Content Distribution

TV Show 27.8%

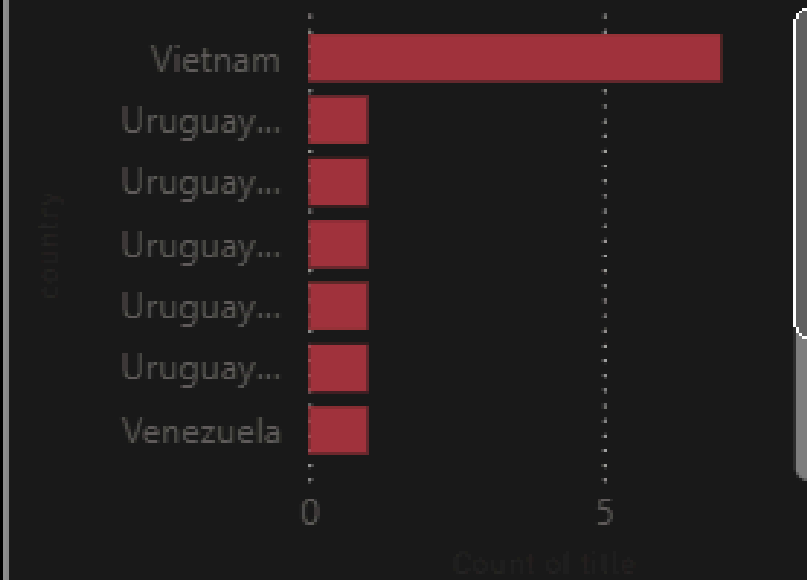


Movie 72.2%

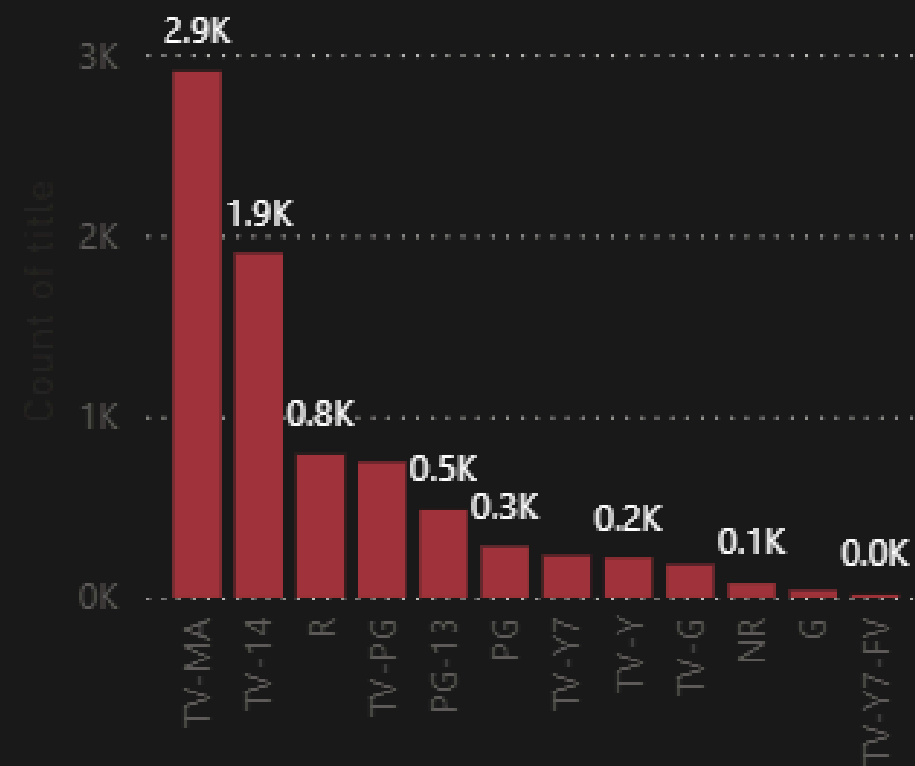
Top 10 Content Producing Countries



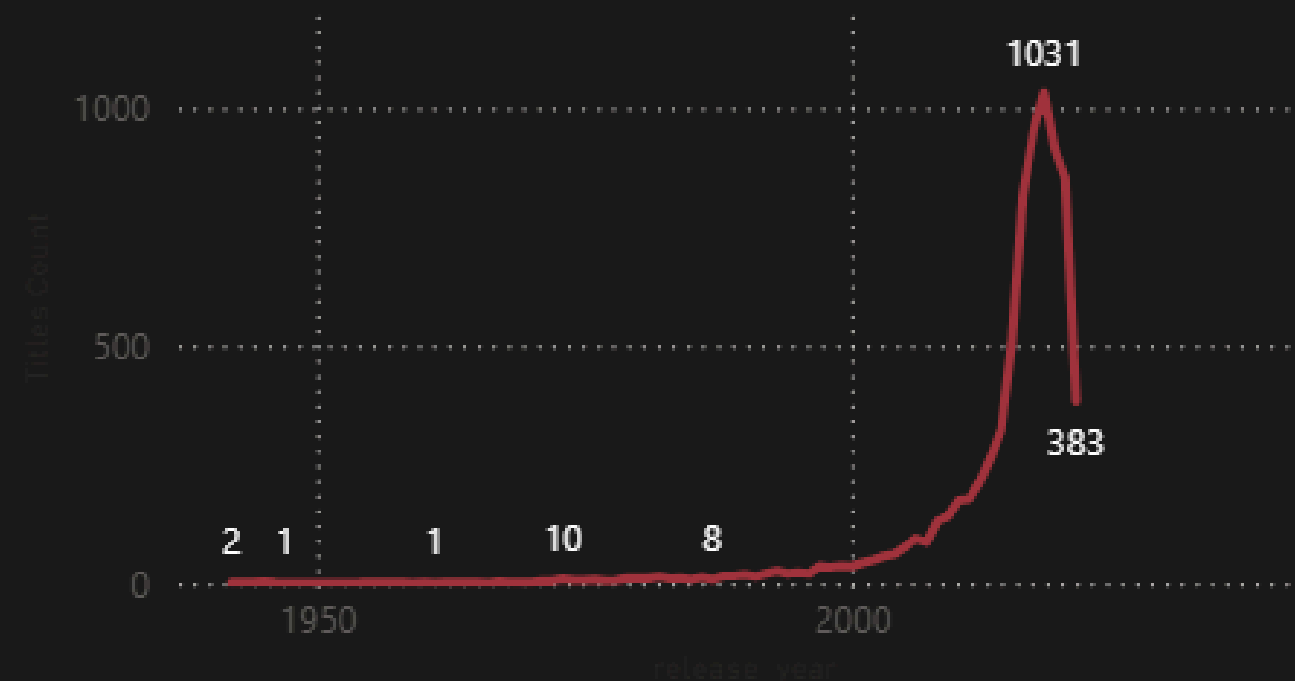
Top 10 Content Producing Countries



Content Ratings Analysis



Content Growth Over Time



Popular Genres



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CONTENT ANALYTICS



OBJECTIVE

To perform exploratory data analysis on the Netflix dataset and identify trends, patterns, and content distribution using Python.

TOOLS & LIBRARIES

- Python
- Pandas
- Matplotlib

ACTIVITIES PERFORMED

- Loaded dataset using Pandas
- Explored dataset structure
- Analyzed missing values
- Performed content distribution analysis
- Created visualizations for business insights



KEY INSIGHTS FROM PYTHON ANALYSIS



Movies dominate Netflix's content library.



TV-MA is the most common content rating.



Content production increased significantly after 2015.



The United States contributes the largest number of titles.



Python enabled efficient data exploration and visualization.



PYTHON DATA ANALYSIS



DATASET OVERVIEW

Dataset Info (netflix_titles.csv)

RangeIndex: 8807 entries, 0 to 8806

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	show_id	8807 non-null	object
1	type	8807 non-null	object
2	title	8807 non-null	object
3	director	6173 non-null	object
4	cast	7982 non-null	object
5	country	7976 non-null	object
6	date_added	8797 non-null	object
7	release_year	8807 non-null	int64
8	rating	8807 non-null	object
9	duration	8807 non-null	object
10	listed_in	8807 non-null	object
11	description	8807 non-null	object

dtypes: int64(1), object(11)

memory usage: 825.8+ KB



MISSING VALUES ANALYSIS

Column

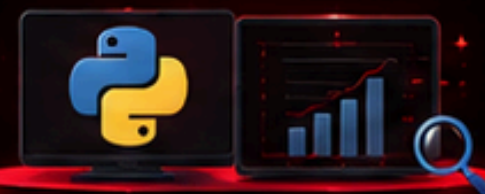
Missing Values

director	2,634
cast	825
country	831
date_added	110
release_year	0
rating	0
duration	0
listed_in	0
description	0
dtype	0



KEY TAKEAWAY

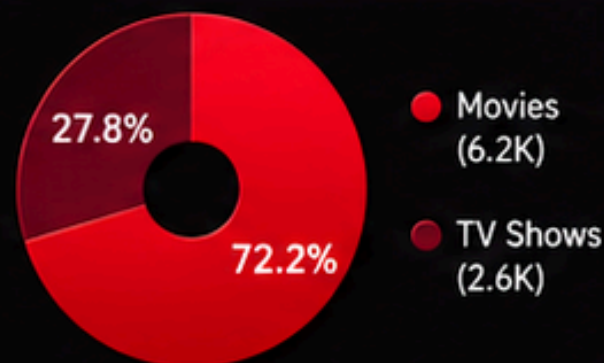
Columns like director, cast and country contain missing values, which were handled during analysis.



PYTHON VISUALIZATIONS & INSIGHTS



1. MOVIES VS TV SHOWS

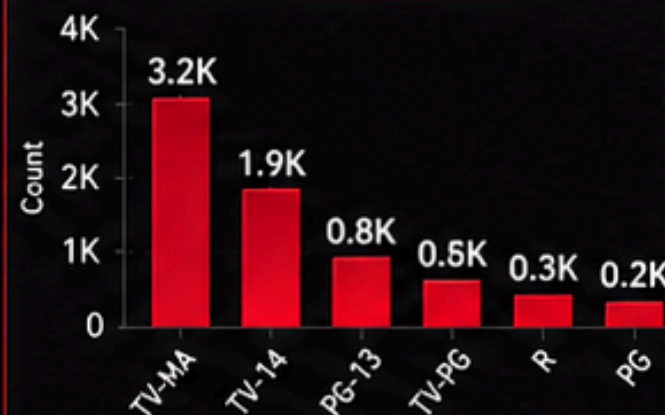


INSIGHT

Movies account for the majority of Netflix content.



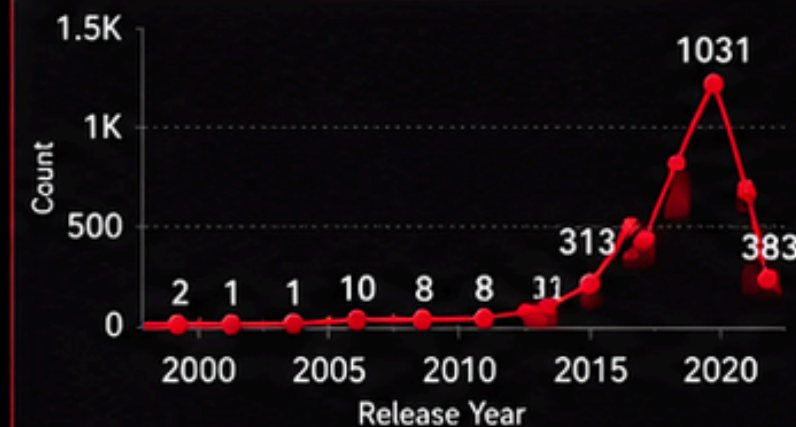
2. TOP RATINGS



INSIGHT

TV-MA is the most frequent content rating.

3. CONTENT RELEASE TREND

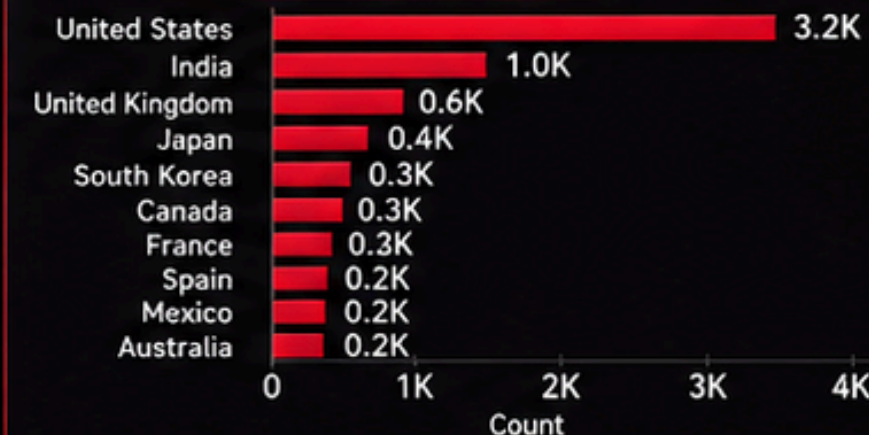


INSIGHT

Content growth accelerated significantly after 2015.



4. TOP COUNTRIES



INSIGHT

The United States contributes the highest number of titles.

KEY FINDINGS & RECOMMENDATIONS



KEY FINDINGS



Movies dominate Netflix content.

Movies account for the majority of the content library.



TV-MA is the most common rating.

TV-MA is the most frequently used content rating on Netflix.



Content growth accelerated after 2015.

Netflix witnessed a significant increase in content addition after 2015.



The United States contributes the most titles.

The United States has the highest number of titles on Netflix.



Diverse genres and global reach.

Netflix offers a wide variety of genres and content from countries around the world.



RECOMMENDATIONS



Invest in high-performing genres.

Continue focusing on popular genres like International Movies, Dramas, and Documentaries.



Expand international and regional content.

Increase content production and licensing from diverse countries to strengthen global reach.



Increase original content in emerging markets.

Focus on creating more original content tailored to local audiences.



Maintain a balanced content mix.

Ensure a healthy mix of movies and TV shows across all age groups and ratings.



Use data-driven insights for decision making.

Leverage data analytics regularly to guide content strategy and business decisions.



This project demonstrates the complete **data analytics workflow**, from data cleaning and exploration to business insights and dashboard development. By leveraging Excel, SQL, Python, and Power BI, we discovered **meaningful patterns and trends** within Netflix's content library, showcasing practical data analytics skills and business intelligence techniques.



DATA INTEGRATION

Integrated and cleaned data using Excel, performed deep analysis using SQL and Python, and visualized insights with Power BI.



INSIGHTFUL ANALYSIS

Identified key trends in content type, ratings, genres, countries, and content growth over time.



BUSINESS VALUE

Insights help understand Netflix's content strategy, audience preferences, and global content distribution.



DATA-DRIVEN DECISIONS

The analysis supports data-driven decision making for content acquisition, production, and audience targeting.



PROJECT IMPACT

This end-to-end project strengthens analytical thinking, problem-solving, and storytelling with data.



OVERALL TAKEAWAY

This project showcases how data analysis can transform raw data into actionable insights, driving better content strategies and business growth for Netflix.





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CONTENT ANALYTICS

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THANK YOU

Thank you for taking the time to review this project.



Netflix Content Analytics Project

End-to-End Data Analytics Using Excel, SQL, Python & Power BI

PREPARED BY



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“ Turning *Data* into *Insights*, and *Insights* into *Decisions*. ”